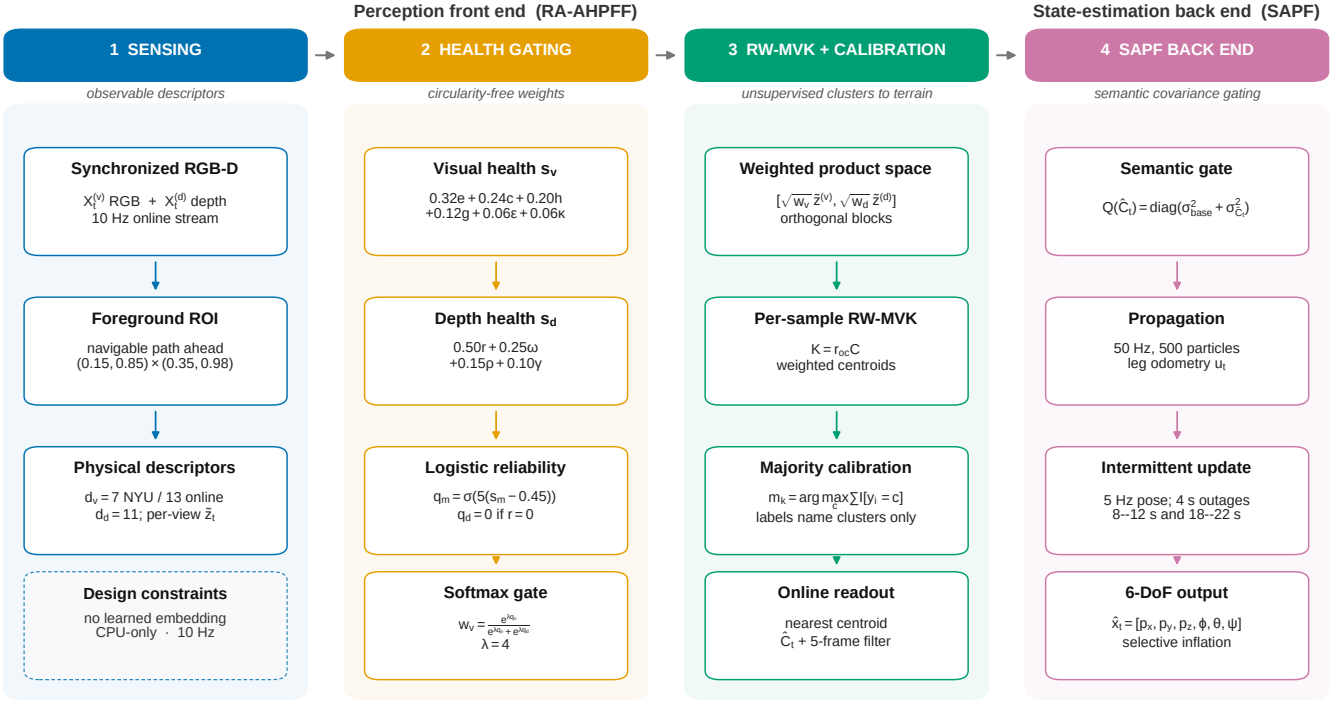
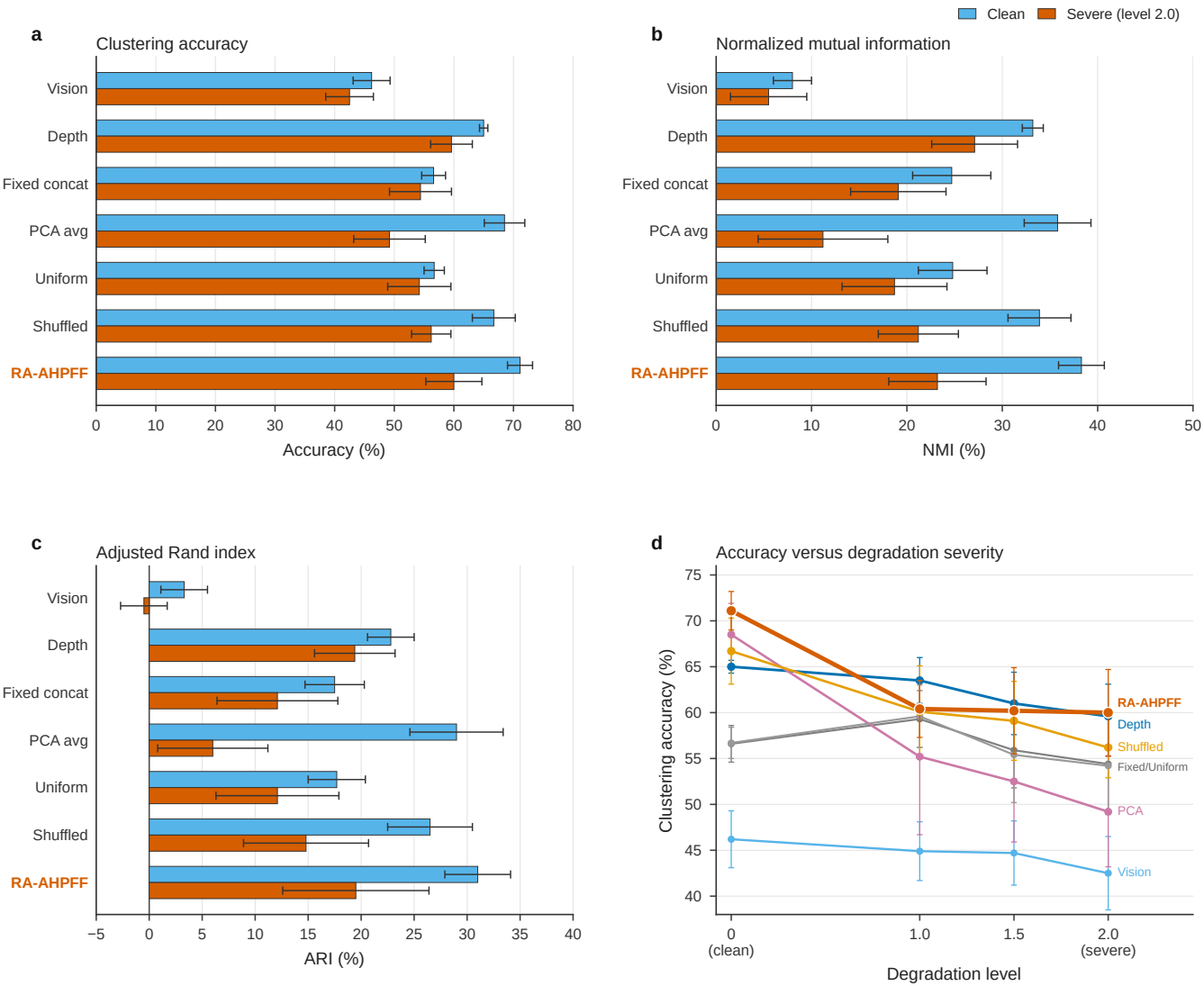


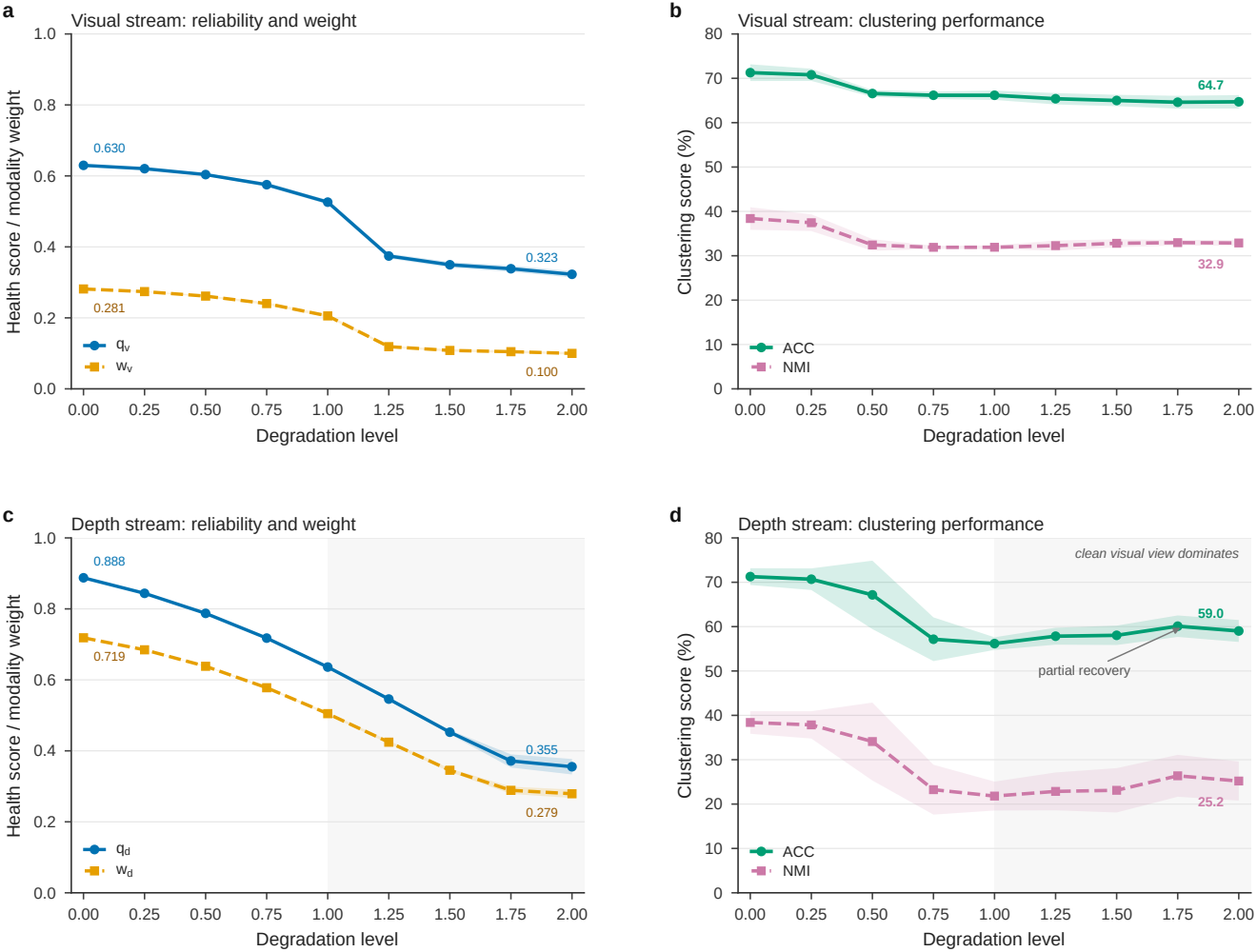
Circularity-free

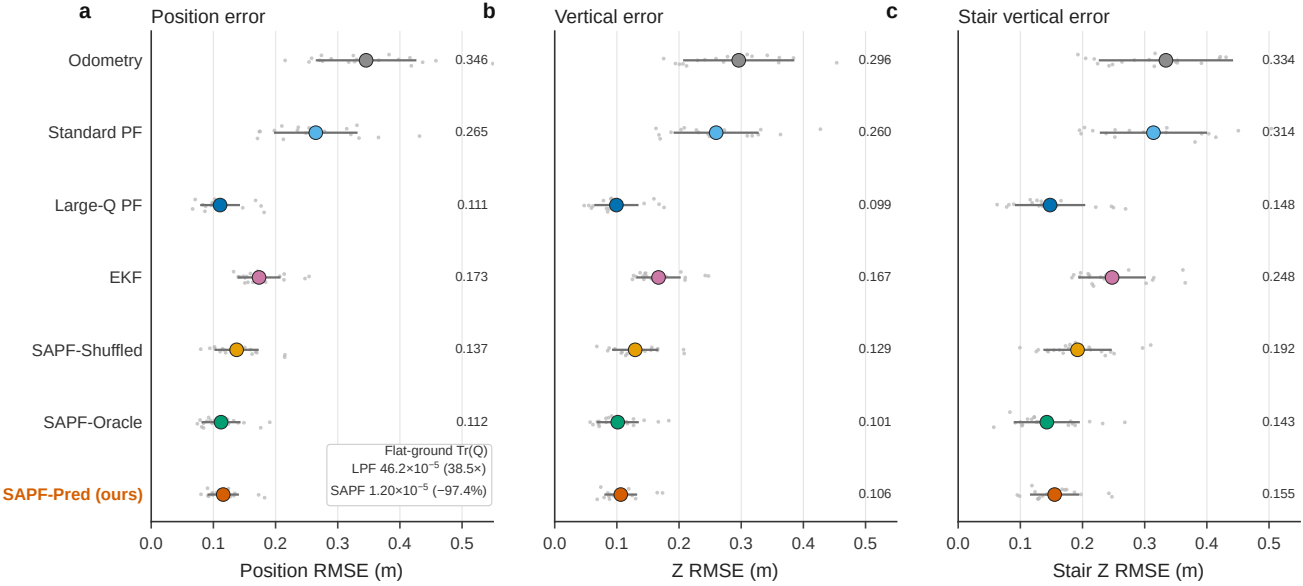
Weights come from observable sensor health before clustering; PCA coordinates are never added across sensor spaces.

Selective gating

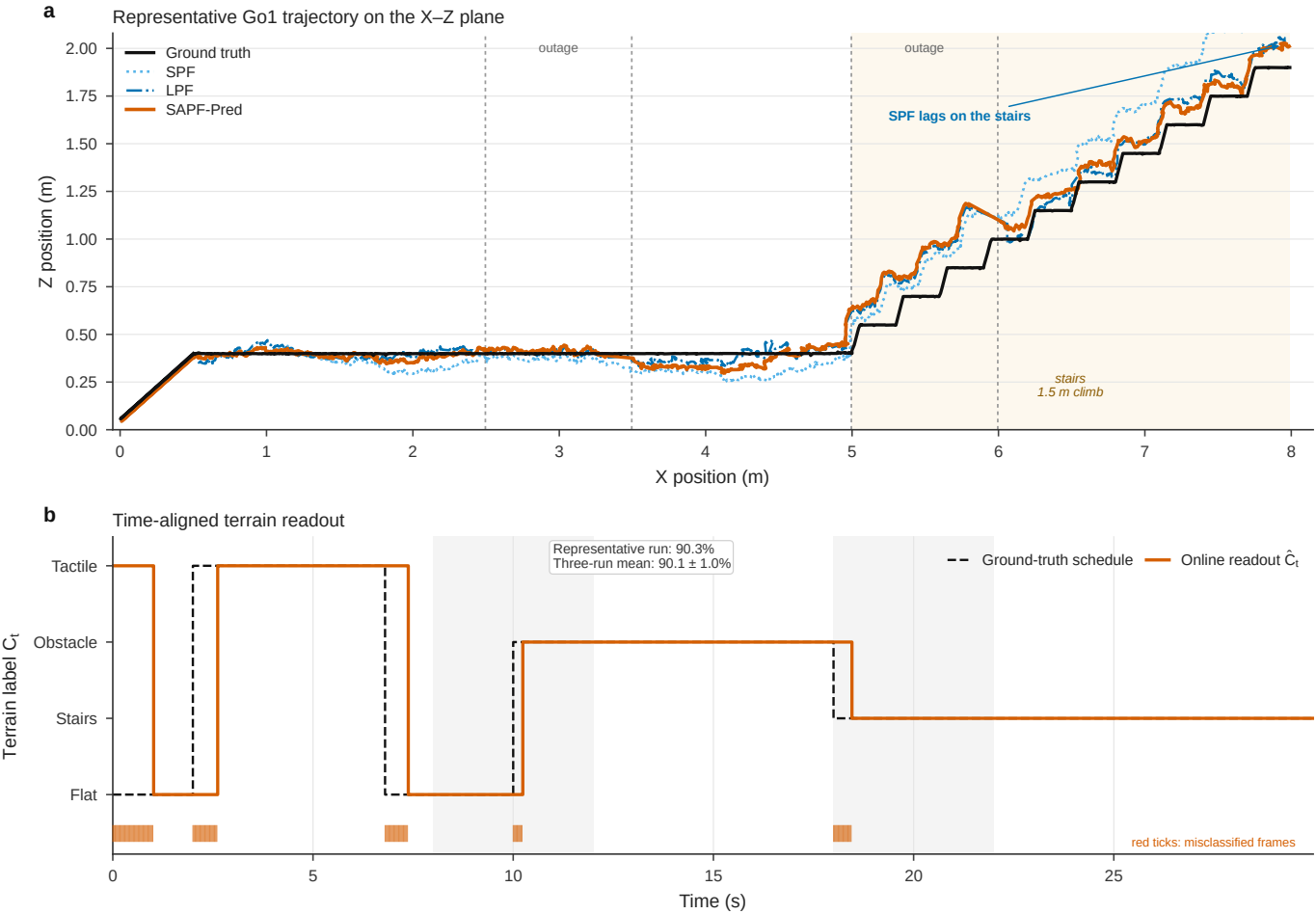
Matches stair tracking while cutting flat-ground process variance by 97.4% (LPF inflates it 38.5x).

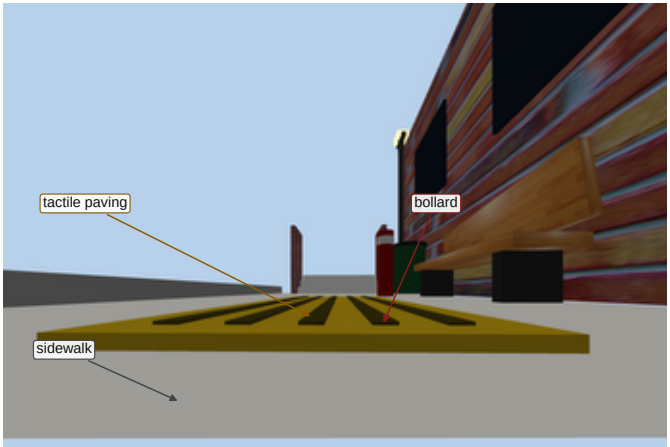

a
Clustering accuracy
Vision
Depth
Fixed concat
PCA avg
Uniform
Shuffled
RA-AHPFF
0 10 20 30 40 50 60 70 80
Accuracy (%)
b
Normalized mutual information
Clean
Severe (level 2.0)
Vision
Depth
Fixed concat
PCA avg
Uniform
Shuffled
RA-AHPFF
0 10 20 30 40 50
NMI (%)
c
Adjusted Rand index
Vision
Depth
Fixed concat
PCA avg
Uniform
Shuffled
RA-AHPFF
−5 0 5 10 15 20 25 30 35 40
ARI (%)
d
Accuracy versus degradation severity
Clustering accuracy (%)
75 70 65 60 55 50 45 40
0
(clean)
1.0
1.5
2.0
(severe)
Degradation level
RA-AHPFF
Depth
Shuffled
Fixed/Uniform
PCA
Vision

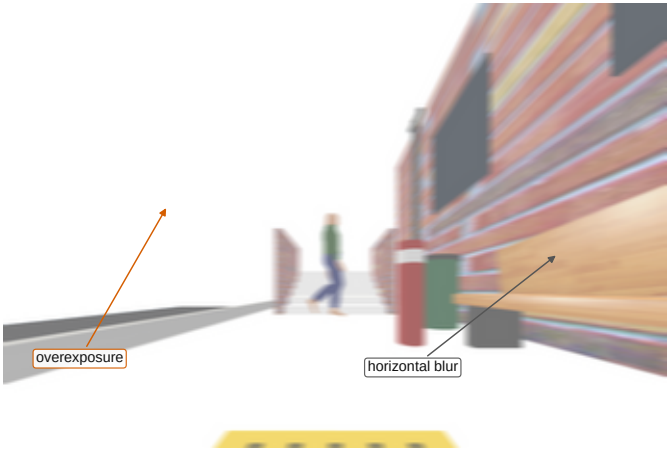

a
Visual stream: reliability and weight
Health score / modality weight
Degradation level
0.630
0.323
0.281
0.100
q_v
w_v
b
Visual stream: clustering performance
Clustering score (%)
Degradation level
64.7
32.9
ACC
NMI
c
Depth stream: reliability and weight
Health score / modality weight
Degradation level
0.888
0.719
0.355
0.279
q_d
w_d
d
Depth stream: clustering performance
Clustering score (%)
Degradation level
clean visual view dominates
59.0
partial recovery
25.2
ACC
NMI



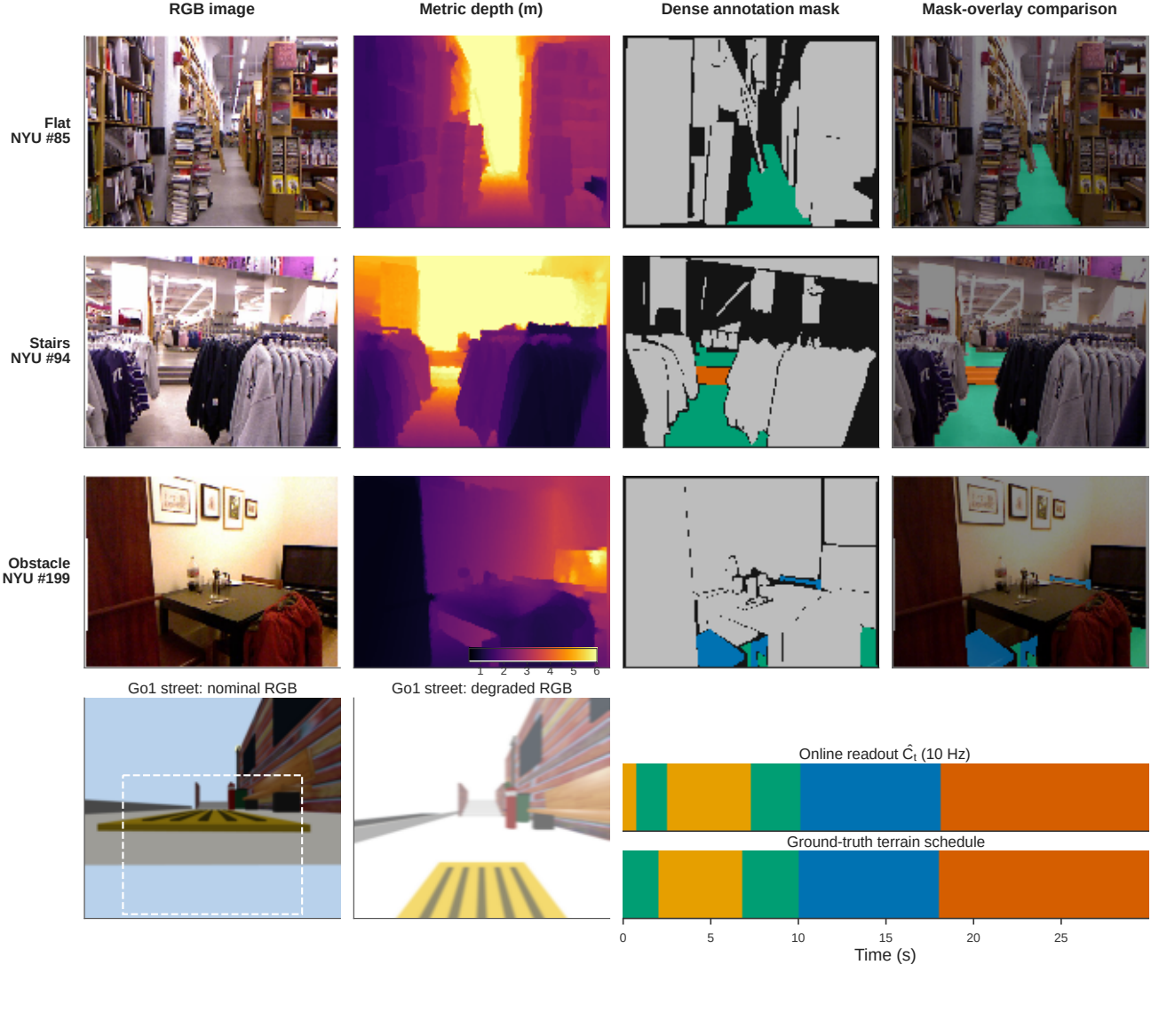
Dots: individual seeds (n = 20). Bar: mean ± s.d. The proposed filter is highlighted.


a
Representative Go1 trajectory on the X–Z plane
Ground truth
SPF
LPF
SAPF-Pred
outage
outage
SPF lags on the stairs
stairs
1.5 m climb
Z position (m)
X position (m)
b
Time-aligned terrain readout
Representative run: 90.3%
Three-run mean: 90.1 ± 1.0%
Ground-truth schedule
Online readout Ĉt
Tactile
Obstacle
Stairs
Flat
Terrain label Ct
red ticks: misclassified frames
Time (s)





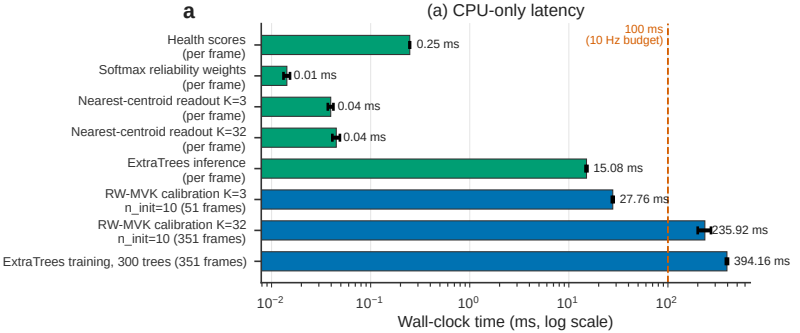
Gazebo Go1 street scene, not a physical-robot photograph. The degraded view applies overexposure and horizontal motion blur; the depth stream loses 50% of its pixels during the same window.

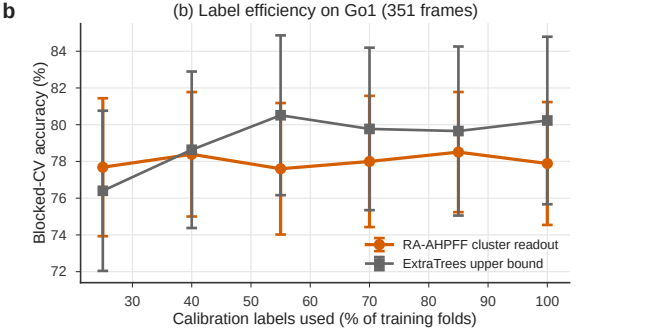


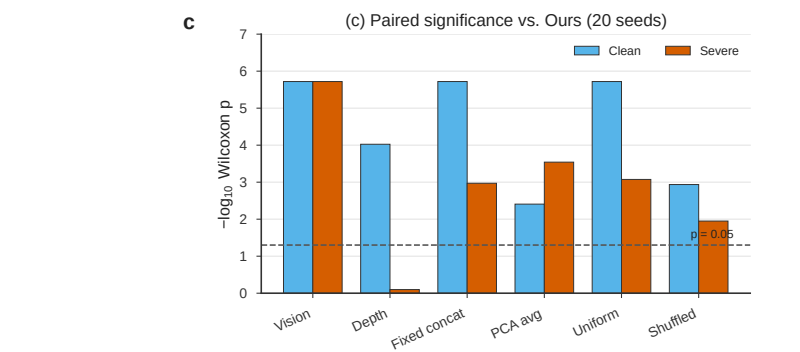
NYU Depth V2 verified subset (17 frames per class; 51 frames). Depth is shown in metres; grey = invalid depth. Dense masks use the class colour code defined in the legend.

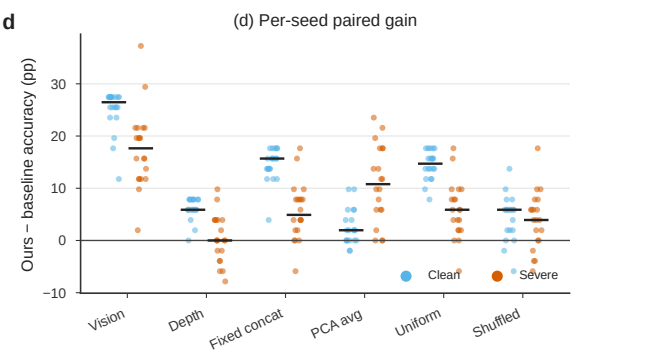


Common t-SNE maps of the clean and severely degraded NYU representations. Colour = predicted terrain, shape = true terrain; shape/colour mismatches are clustering errors. Severe corruption: each sample loses exactly one modality (level 2.0)

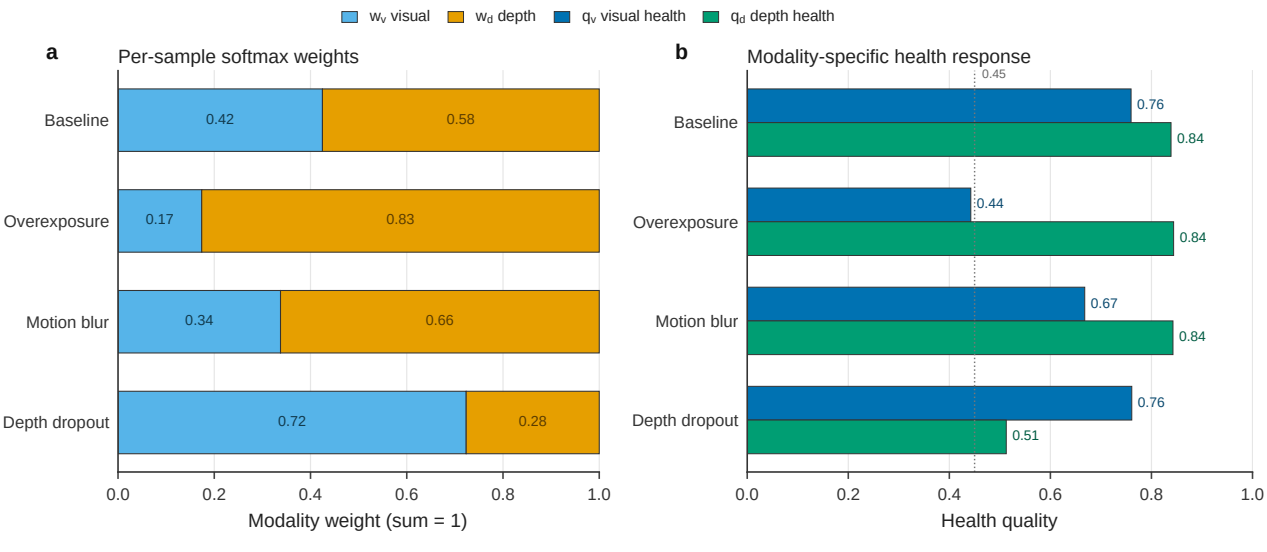




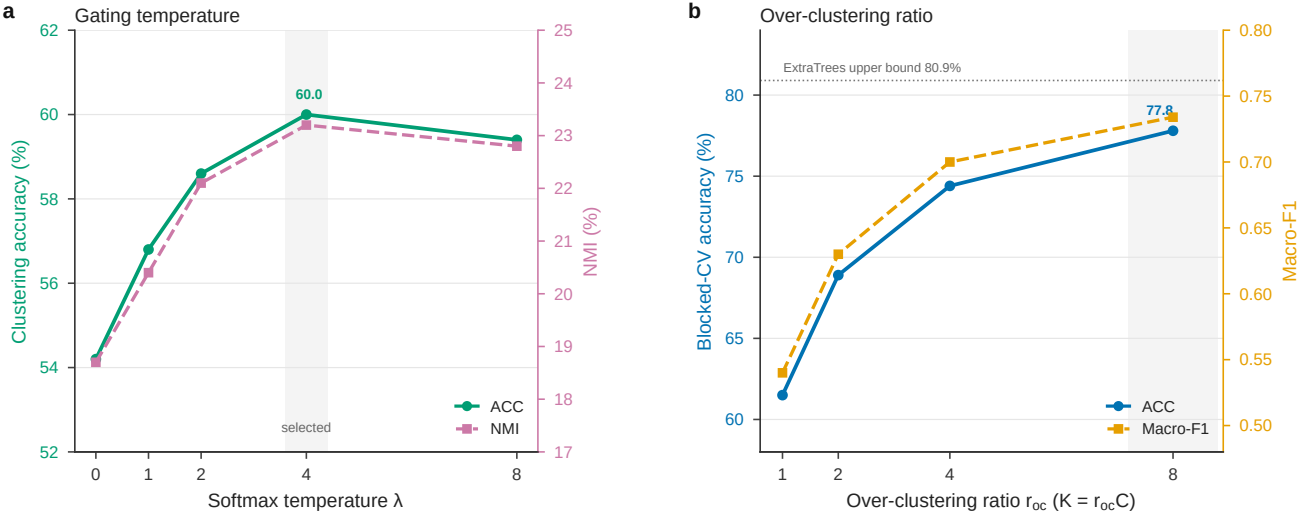


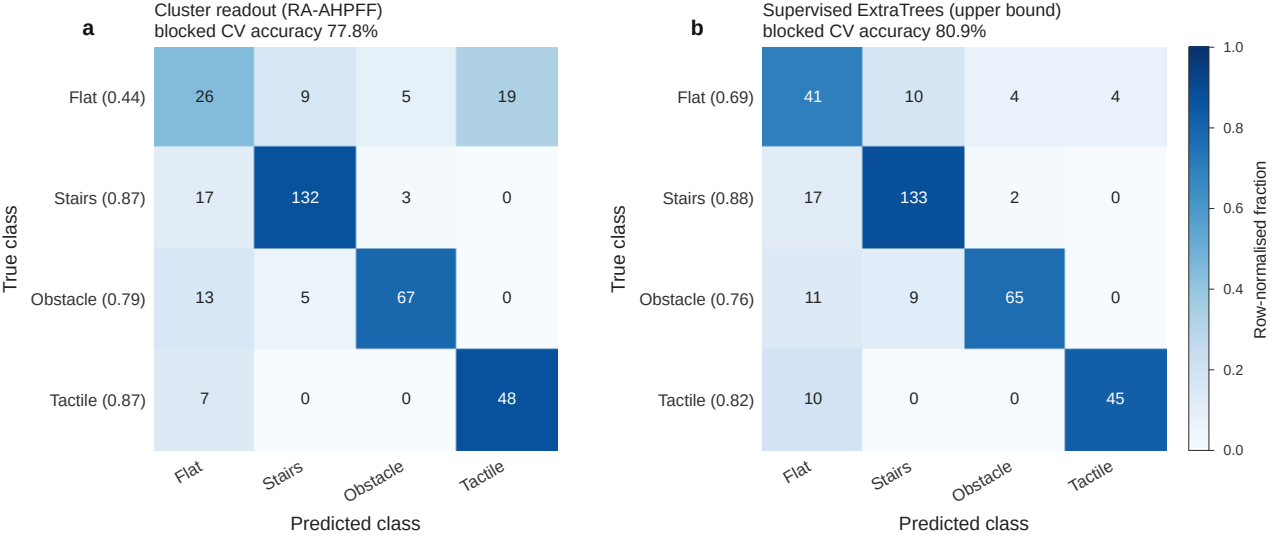


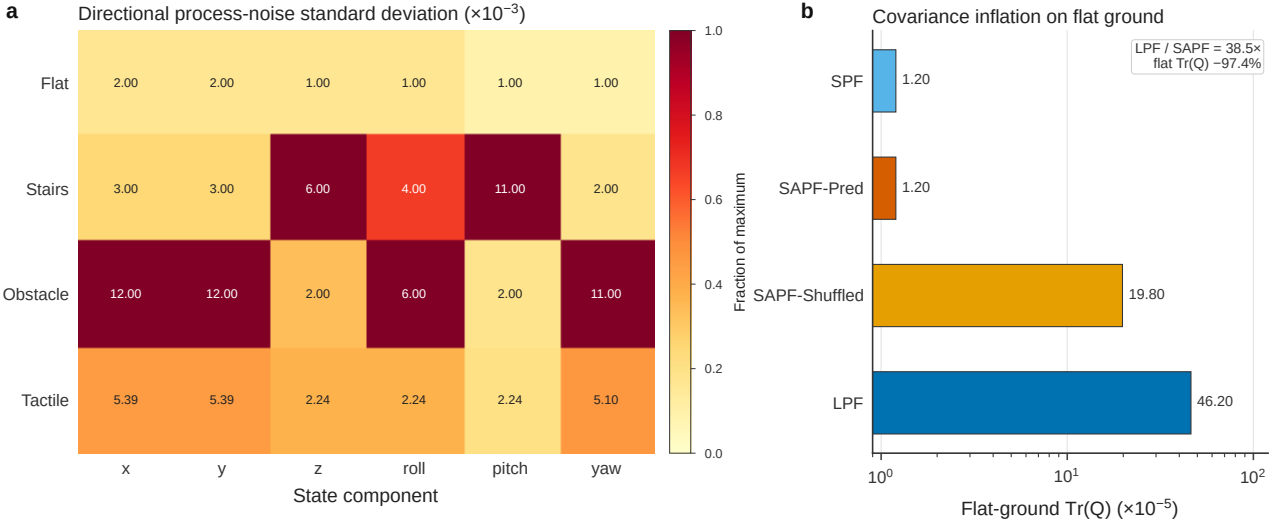
All timings are single-thread CPU measurements (Python 3.8, 2.4 GHz class x86-64). The 10 Hz perception budget is 100 ms per RGB-D frame; calibration is an offline in-scene step. Panel (b) error bars are s.e.m. over 25 blocked evaluations.



Corruption active during 6–20 s in the Go1 street scene; weights are computed per frame, not set globally.


a
Gating temperature
Clustering accuracy (%)
NMI (%)
60.0
selected
ACC
NMI
Softmax temperature λ
b
Over-clustering ratio
ExtraTrees upper bound 80.9%
77.8
Blocked-CV accuracy (%)
Macro-F1
ACC
Macro-F1
Over-clustering ratio r_oc (K = r_ocC)


a
Cluster readout (RA-AHPFF)
blocked CV accuracy 77.8%
Flat (0.44)
Stairs (0.87)
Obstacle (0.79)
Tactile (0.87)
26
9
5
19
17
132
3
0
13
5
67
0
7
0
0
48
Flat
Stairs
Obstacle
Tactile
True class
Predicted class
b
Supervised ExtraTrees (upper bound)
blocked CV accuracy 80.9%
Flat (0.69)
Stairs (0.88)
Obstacle (0.76)
Tactile (0.82)
41
10
4
4
17
133
2
0
11
9
65
0
10
0
0
45
Flat
Stairs
Obstacle
Tactile
True class
Predicted class
Row-normalised fraction
1.0
0.8
0.6
0.4
0.2
0.0
Temporally blocked 5-fold CV on 351 Go1 samples (K = 32, r_{oc} = 8); chance = 25%.


a
Directional process-noise standard deviation (×10⁻³)
Flat
Stairs
Obstacle
Tactile
2.00 2.00 1.00 1.00 1.00 1.00
3.00 3.00 6.00 4.00 11.00 2.00
12.00 12.00 2.00 6.00 2.00 11.00
5.39 5.39 2.24 2.24 2.24 5.10
x
y
z
roll
pitch
yaw
State component
Fraction of maximum
1.0
0.8
0.6
0.4
0.2
0.0
b
Covariance inflation on flat ground
LPF / SAPF = 38.5×
flat Tr(Q) −97.4%
SPF
SAPF-Pred
SAPF-Shuffled
LPF
1.20
1.20
19.80
46.20
10⁰
10¹
10²
Flat-ground Tr(Q) (×10⁻⁵)